

Macro Three Replication Project: Investigating the Economic Impact of the CFA Franc Zone with Synthetic Control Methods

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Test

1 Replication Target

In this project, we aim to replicate the findings of *From the French franc to the euro, is there an economic impact for the CFA franc zone countries? A 'bias-corrected' synthetic control method* by Alassane Diallo and Adama Ba, published in Cogent Economics & Finance in 2024. This paper assesses the impact of pegging the CFA franc to the euro on real GDP per capita in the West African Economic and Monetary Union (WAEMU) and the Central African Economic and Monetary Community (CAEMC) countries. The original study uses a set of 35 annual country-level panel data from 1980 to 2019 and applies synthetic control methods to estimate the counterfactual economic outcomes for member countries.

2 Data

Table 1: Variables

Variable	Type	Category	(First) Page Ref	Possible Sources
Real GDP per capita	Continuous	Outcome	6	WDI, PennWorld Tables
Lagged GDP per capita	Continuous	Lagged outcome	10	WDI, PennWorld Tables
Gross Fixed Capital Formation (% GDP)	Continuous	Capital	10	
Government expenditure (% GDP)	Continuous	Fiscal	10	WDI, PennWorld Tables
Official Development Assistance (% GDP)	Continuous	External	10	Source not listed
Foreign Development Investment (% GDP)	Continuous	External	10	WDI
Employment rate	Continuous	Labour	10	WDI
Agriculture (% GDP)	Continuous	Structure	10	WDI, PennWorld Tables
Industry (% GDP)	Continuous	Structure	10	WDI, PennWorld Tables
Governance / institutional quality	Index	Institutions	10	Polity5 Dataset
Political regime characteristics	Categorical/Index	Institutions	10	Polity5 Dataset
Inflation	Continuous	Macro controls	5	WDI, PennWorld Tables
Euro peg dummy	Binary	Treatment		Authors
Synthetic GDP per capita	$\bar{Y}_{it}$	Continuous	Constructed	
Treatment effect	τ_{it}	Continuous	Constructed	

Table 2: CFA franc zone countries annual inflation rate: WAEMU

country	1980–2001	2002–2021
Benin	7.5270	2.1268
Burkina Faso	4.3427	2.2660
Côte d'Ivoire	8.5558	2.1842
Guinea-Bissau	39.2343	2.8921
Mali	8.6642	3.5490
Niger	6.0235	2.8630
Senegal	5.1354	1.8448
Togo	6.2716	2.0069
Average inflation for the WAEMU zone	10.7193	2.4666
Average inflation without Guinea-Bissau	6.6457	2.4058

Table 3: CFA franc zone countries annual inflation rate: CAEMC

country	1980–2001	2002–2021
Cameroon	7.0840	1.8268
Central African Republic	6.5481	2.9430
Chad	5.2621	5.4660
Republic of Congo	7.3655	6.1262
Equatorial Guinea	11.0392	6.7586
Gabon	7.4425	3.7486
Average inflation for the CAEMC zone	7.4569	4.4782

Table 4: Non-CFA franc zone countries mean annual inflation rate

country	1980–2001	2002–2021
Angola	550.9314	32.1383
Burundi	10.3255	9.2785
Congo, Dem. Rep.	1810.0588	16.3427
Ethiopia	5.2079	13.5134
Gambia, The	14.5303	10.1870
Ghana	36.1786	21.0285
Guinea	16.1342	11.2597
Kenya	11.8579	8.5152
Madagascar	18.4232	8.3332
Malawi	23.6956	18.1634
Nigeria	30.4076	13.5749
Sao Tome and Principe	31.6514	10.6113
Sierra Leone	51.3708	11.1235
Sudan	50.6410	37.8117
Tanzania	21.8489	7.4472
Uganda	53.1089	8.7408
Zambia	51.1202	12.4355
Zimbabwe	-1.6381	71.6617
Average inflation	154.7697	17.8981

Table 5: WAEMU countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Benin	1.0235	1.0352	1.0665	1.0540
Burkina Faso	1.0213	1.0213	1.0066	1.0678
Cote d'Ivoire	0.9847	1.0481	1.0273	1.0522
Guinea-Bissau	1.0112	1.0037	0.9620	1.0572
Mali	1.0615	1.0303	1.0187	1.0586
Niger	0.9934	1.0067	0.9991	1.0587
Senegal	0.9983	1.0078	0.9863	1.0502
Togo	0.9903	1.0427	0.9923	1.0481
Average for the WAEMU area	1.0105	1.0245	1.0074	1.0559
Average without Guinea-Bissau	1.0104	1.0274	1.0138	1.0557

Table 6: CAEMC countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Cameroon	0.9961	1.0178	0.9911	1.0467
Central African Republic	0.9868	1.0025	0.9609	1.0432
Chad	0.9908	1.0347	0.9916	1.1043
Congo, Rep.	1.0200	1.0522	1.0029	1.0729
Equatorial Guinea	1.2073	1.0537	1.2602	1.1028
Gabon	1.0071	1.0372	1.0011	1.0490
Average for the CAEMC zone	1.0347	1.0330	1.0346	1.0698

Table 7: Non-CFA countries' mean annual GDP per capita growth factor: PWT and WDI GDP variables

country	gdp_pc 1990-2001	gdp_pc 2002-2021	gdp_pc_wdi 1990-2001	gdp_pc_wdi 2002-2021
Angola	0.9892	1.0757	0.9813	1.1195
Burundi	0.9770	1.0039	0.9675	1.0379
Congo, Dem. Rep.	0.9078	1.0381	1.1332	1.0809
Ethiopia	0.9954	1.0839	0.9449	1.1115
Gambia, The	1.0012	0.9921	1.0723	1.0453
Ghana	1.0087	1.0323	0.9871	1.1360
Guinea	0.9770	0.9955	0.9886	1.0558
Kenya	0.9972	1.0383	1.0209	1.0868
Madagascar	0.9942	1.0106	1.0232	1.0281
Malawi	0.9803	1.0229	1.0217	1.0738
Nigeria	1.0205	1.1040	1.0919	1.0981
Sao Tome and Principe	1.0179	1.0284	0.9814	1.0823
Sierra Leone	0.9870	1.0315	1.0978	1.0515
Sudan	1.0116	1.0402	1.0677	1.0388
Tanzania	1.0144	1.0372	1.0406	1.0570
Uganda	1.0358	1.0264	0.9921	1.0776
Zambia	0.9996	1.0526	0.9827	1.0701
Zimbabwe	0.9869	1.0036	0.9753	1.1332
Average	0.9945	1.0343	1.0206	1.0769

Table 8: WAEMU countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Benin	0	0	0	0
Burkina Faso	0	0	0	0
Côte d'Ivoire	0	0	0	0
Guinea-Bissau	0	0	0	0
Mali	0	0	0	0
Niger	0	0	0	0
Senegal	0	0	0	0
Togo	0	0	0	0
Average for the WAEMU area	0	0	0	0
Average without Guinea-Bissau	0	0	0	0

Table 9: CAEMC countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Cameroon	0	0	0	0
Central African Republic	0	0	0	0
Chad	0	0	0	0
Republic of Congo	0	0	0	0
Equatorial Guinea	0	0	0	0
Gabon	0	0	0	0
Average for the CAEMC zone	0	0	0	0

Table 10: Non-CFA countries' mean annual GDP per capita growth: WDI constant and current GDP per capita

country	constant GDP pc growth 1990-2001	constant GDP pc growth 2002-2021	current GDP pc growth 1990-2001	current GDP pc growth 2002-2021
Angola	0	0	0	0
Burundi	0	0	0	0
Congo, Dem. Rep.	0	0	0	0
Ethiopia	0	0	0	0
Gambia, The	0	0	0	0
Ghana	0	0	0	0
Guinea	0	0	0	0
Kenya	0	0	0	0
Madagascar	0	0	0	0
Malawi	0	0	0	0
Nigeria	0	0	0	0
Sao Tome and Principe	0	0	0	0
Sierra Leone	0	0	0	0
Sudan	0	0	0	0
Tanzania	0	0	0	0
Uganda	0	0	0	0
Zambia	0	0	0	0
Zimbabwe	0	0	0	0
Average	0	0	0	0

Table 11: GDP method comparison for AGO

country	method	1980-2021	1990-2001	2002-2021	n 1980-2021	n 1990-2001	n 2002-2021
Angola	growth_factor	1.0244	0.9853	1.0717	39	12	18
Angola	index_change_from_1990	40.6359	-9.2907	91.5151	40	12	18
Angola	log_growth	1.6480	-2.1830	6.0821	39	12	18
Angola	pct_growth	2.4407	-1.4653	7.1684	39	12	18
Angola	growth_factor	1.0250	0.9865	1.0723	39	12	18
Angola	index_change_from_1990	41.5124	-15.1002	99.3930	40	12	18
Angola	log_growth	1.6146	-2.2800	6.0638	39	12	18
Angola	pct_growth	2.5030	-1.3499	7.2327	39	12	18
Angola	growth_factor	1.0254	0.9889	1.0712	39	12	18
Angola	index_change_from_1990	42.8099	-6.8674	95.2394	40	12	18
Angola	log_growth	1.7364	-1.8919	6.0563	39	12	18
Angola	pct_growth	2.5383	-1.1105	7.1167	39	12	18
Angola	growth_factor	1.0031	0.9848	1.0206	39	12	18
Angola	index_change_from_1990	5.6979	-18.6092	23.0705	40	12	18
Angola	log_growth	0.0686	-2.0153	1.8884	39	12	18
Angola	pct_growth	0.3120	-1.5197	2.0564	39	12	18
Angola	growth_factor	1.0275	0.9892	1.0757	39	12	18
Angola	index_change_from_1990	48.4202	-11.9331	113.6706	40	12	18
Angola	log_growth	1.8587	-1.7830	6.2262	39	12	18
Angola	pct_growth	2.7523	-1.0796	7.5692	39	12	18
Angola	growth_factor	0.9994	0.9826	1.0143	41	12	20
Angola	index_change_from_1990	6.4294	-18.7752	20.1983	42	12	20
Angola	log_growth	-0.2871	-2.2457	1.2887	41	12	20
Angola	pct_growth	-0.0573	-1.7407	1.4317	41	12	20
Angola	growth_factor	1.0591	0.9813	1.1195	41	12	20
Angola	index_change_from_1990	93.6779	-42.3873	234.5528	42	12	20
Angola	log_growth	2.8062	-4.4279	7.3137	41	12	20
Angola	pct_growth	5.9118	-1.8725	11.9469	41	12	20
Angola	growth_factor	1.0047	0.9871	1.0143	31	11	20
Angola	index_change_from_1990	5.5832	-18.7752	20.1983	32	12	20
Angola	log_growth	0.1848	-1.8222	1.2887	31	11	20
Angola	pct_growth	0.4650	-1.2926	1.4317	31	11	20
Angola	growth_factor	1.0320	1.0075	1.0455	31	11	20
Angola	index_change_from_1990	55.9656	-8.3084	94.5257	32	12	20
Angola	log_growth	2.8134	0.2344	4.2319	31	11	20
Angola	pct_growth	3.1977	0.7450	4.5467	31	11	20
Angola	growth_factor	1.0026	0.9832	1.0221	39	12	18

Table 12: GDP method comparison for BEN

country	method	1980-2021	1990-2001	2002-2021	n 1980-2021	n 1990-2001	n 2002-2021
Benin	growth_factor	1.0149	1.0214	1.0376	39	12	18
Benin	index_change_from_1990	43.8532	5.3929	79.1517	40	12	18
Benin	log_growth	1.2244	2.0054	3.6223	39	12	18
Benin	pct_growth	1.4889	2.1420	3.7590	39	12	18
Benin	growth_factor	1.0154	1.0238	1.0366	39	12	18
Benin	index_change_from_1990	48.2725	6.4867	83.8784	40	12	18
Benin	log_growth	1.1674	2.2417	3.5315	39	12	18
Benin	pct_growth	1.5423	2.3778	3.6636	39	12	18
Benin	growth_factor	1.0142	1.0215	1.0369	39	12	18
Benin	index_change_from_1990	44.3284	6.0907	78.3218	40	12	18
Benin	log_growth	1.1555	2.0165	3.5513	39	12	18
Benin	pct_growth	1.4248	2.1515	3.6907	39	12	18
Benin	growth_factor	1.0234	1.0207	1.0346	39	12	18
Benin	index_change_from_1990	26.9422	8.2695	55.4429	40	12	18
Benin	log_growth	2.2432	2.0303	3.3362	39	12	18
Benin	pct_growth	2.3379	2.0736	3.4587	39	12	18
Benin	growth_factor	1.0136	1.0235	1.0352	39	12	18
Benin	index_change_from_1990	48.3190	6.6976	81.2933	40	12	18
Benin	log_growth	0.9802	2.2079	3.3902	39	12	18
Benin	pct_growth	1.3648	2.3455	3.5206	39	12	18
Benin	growth_factor	1.0120	1.0183	1.0148	41	12	20
Benin	index_change_from_1990	16.7945	6.4319	30.9259	42	12	20
Benin	log_growth	1.1521	1.7970	1.4522	41	12	20
Benin	pct_growth	1.1984	1.8291	1.4789	41	12	20
Benin	growth_factor	1.0425	1.0665	1.0540	41	12	20
Benin	index_change_from_1990	76.2245	3.6541	168.3044	42	12	20
Benin	log_growth	3.1801	4.3169	4.9254	41	12	20
Benin	pct_growth	4.2485	6.6491	5.3989	41	12	20
Benin	growth_factor	1.0148	1.0148	1.0148	31	11	20
Benin	index_change_from_1990	21.7406	6.4319	30.9259	32	12	20
Benin	log_growth	1.4543	1.4582	1.4522	31	11	20
Benin	pct_growth	1.4790	1.4792	1.4789	31	11	20
Benin	growth_factor	1.0353	1.0359	1.0350	31	11	20
Benin	index_change_from_1990	75.4385	20.9206	108.1456	32	12	20
Benin	log_growth	3.4575	3.5148	3.4259	31	11	20
Benin	pct_growth	3.5346	3.5885	3.5050	31	11	20
Benin	growth_factor	1.0117	1.0174	1.0147	39	12	18

Table 13: GDP method comparison for CMR

country	method	1980-2021	1990-2001	2002-2021	n 1980-2021	n 1990-2001	n 2002-2021
Cameroon	growth_factor	1.0107	0.9960	1.0167	39	12	18
Cameroon	index_change_from_1990	11.0875	-2.3555	17.5176	40	12	18
Cameroon	log_growth	0.9822	-0.4636	1.6474	39	12	18
Cameroon	pct_growth	1.0670	-0.4014	1.6687	39	12	18
Cameroon	growth_factor	1.0127	0.9962	1.0181	39	12	18
Cameroon	index_change_from_1990	10.0362	-4.6244	17.8727	40	12	18
Cameroon	log_growth	1.1662	-0.4784	1.7822	39	12	18
Cameroon	pct_growth	1.2736	-0.3777	1.8088	39	12	18
Cameroon	growth_factor	1.0103	0.9960	1.0164	39	12	18
Cameroon	index_change_from_1990	11.5159	-1.9857	17.2610	40	12	18
Cameroon	log_growth	0.9377	-0.4645	1.6145	39	12	18
Cameroon	pct_growth	1.0261	-0.3967	1.6367	39	12	18
Cameroon	growth_factor	0.9993	0.9855	1.0149	39	12	18
Cameroon	index_change_from_1990	6.8061	-11.9187	4.7897	40	12	18
Cameroon	log_growth	-0.1518	-1.5396	1.4764	39	12	18
Cameroon	pct_growth	-0.0746	-1.4470	1.4939	39	12	18
Cameroon	growth_factor	1.0118	0.9961	1.0178	39	12	18
Cameroon	index_change_from_1990	10.6938	-4.4973	17.9228	40	12	18
Cameroon	log_growth	1.0733	-0.4883	1.7501	39	12	18
Cameroon	pct_growth	1.1792	-0.3935	1.7777	39	12	18
Cameroon	growth_factor	1.0034	0.9810	1.0107	41	12	20
Cameroon	index_change_from_1990	-0.7680	-15.6276	-1.3273	42	12	20
Cameroon	log_growth	0.2449	-2.0280	1.0509	41	12	20
Cameroon	pct_growth	0.3407	-1.9032	1.0659	41	12	20
Cameroon	growth_factor	1.0279	0.9911	1.0467	41	12	20
Cameroon	index_change_from_1990	4.4624	-18.8772	28.1761	42	12	20
Cameroon	log_growth	1.8474	-2.7976	4.2452	41	12	20
Cameroon	pct_growth	2.7919	-0.8929	4.6665	41	12	20
Cameroon	growth_factor	1.0024	0.9873	1.0107	31	11	20
Cameroon	index_change_from_1990	-6.6899	-15.6276	-1.3273	32	12	20
Cameroon	log_growth	0.1913	-1.3718	1.0509	31	11	20
Cameroon	pct_growth	0.2359	-1.2733	1.0659	31	11	20
Cameroon	growth_factor	1.0286	1.0076	1.0402	31	11	20
Cameroon	index_change_from_1990	35.6411	-4.7006	59.8485	32	12	20
Cameroon	log_growth	2.7687	0.6849	3.9148	31	11	20
Cameroon	pct_growth	2.8619	0.7648	4.0153	31	11	20
Cameroon	growth_factor	1.0040	0.9808	1.0143	39	12	18

Table 14: GDP table: PWT $pwt_{c}gdp_{p}clog_{g}rowth$

group	country	1980-2021	1990-2001	2002-2021
WAEMU	Average WAEMU	1.0804	0.7616	2.2581
WAEMU	Benin	1.1674	2.2417	3.5315
WAEMU	Burkina Faso	1.5537	1.9199	2.2395
WAEMU	Cote d'Ivoire	1.5994	-1.6068	4.4222
WAEMU	Guinea-Bissau	1.1549	0.5352	0.7114
WAEMU	Mali	3.3211	5.4376	2.9819
WAEMU	Niger	-1.1471	-0.7264	0.7181
WAEMU	Senegal	1.0197	-0.2089	0.8103
WAEMU	Togo	-0.0260	-1.4991	2.6502
CAEMC	Average CAEMC	1.5797	2.0442	2.1020
CAEMC	Cameroon	1.1662	-0.4784	1.7822
CAEMC	Central African Republic	-0.2746	-1.5593	-0.1089
CAEMC	Chad	1.4716	-0.8877	2.0108
CAEMC	Congo, Rep.	1.3622	0.5361	3.2055
CAEMC	Equatorial Guinea	5.5471	14.6762	3.2707
CAEMC	Gabon	0.2058	-0.0220	2.4519
Non-CFA comparison	Angola	1.6146	-2.2800	6.0638
Non-CFA comparison	Average Non-CFA comparison	0.7355	-1.0552	2.9967
Non-CFA comparison	Burundi	-0.2470	-2.2956	0.7036
Non-CFA comparison	Congo, Dem. Rep.	-1.9490	-10.9170	3.7331
Non-CFA comparison	Ethiopia	3.5463	-0.7644	7.8606
Non-CFA comparison	Gambia, The	-0.3137	0.0947	-0.9057
Non-CFA comparison	Ghana	0.9757	0.4745	2.9075
Non-CFA comparison	Guinea	-0.8491	-2.3514	-0.3911
Non-CFA comparison	Kenya	1.7803	-0.3148	3.7649
Non-CFA comparison	Madagascar	0.1204	-0.6210	1.0193
Non-CFA comparison	Malawi	-0.1458	-2.3708	1.7212
Non-CFA comparison	Nigeria	-0.0898	0.0979	8.8654
Non-CFA comparison	Sao Tome and Principe	1.2638	1.6071	2.9151
Non-CFA comparison	Sierra Leone	1.3110	-2.0725	2.9975
Non-CFA comparison	Sudan	1.6204	1.0182	3.5345
Non-CFA comparison	Tanzania	0.7309	0.9827	3.4604
Non-CFA comparison	Uganda	2.4035	3.2411	2.6062
Non-CFA comparison	Zambia	1.7414	-0.3149	4.7613
Non-CFA comparison	Zimbabwe	-0.2753	-2.2073	-1.6776

Table 15: Estimated synthetic control weights by treated country

Donor country	Benin	Burkina F.	Ivory Coast	Mali	Niger	Senegal	Togo	Cameroon	CAR	Chad	Congo	Gabon	E. Guinea
Bangladesh	.976	.375	.680	0	.999	.647	.970	.656	.775	.546	.701	0	0
Barbados	0	0	0	0	0	0	0	0	0	0	0	0	0
Bhutan	0	0	0	0	0	.179	0	0	0	0	0	0	0
Bolivia	0	0	0	0	0	.024	0	.109	0	0	0	0	0
Botswana	0	0	0	0	0	0	0	0	0	0	0	0	.280
Cabo Verde	0	0	0	0	0	.012	0	0	0	0	.070	0	.677
Dominica	0	0	.001	0	0	.077	0	0	0	0	0	0	0
Ecuador	.024	0	.205	0	0	.061	.030	.167	0	0	.190	.485	0
Eswatini	0	0	0	0	0	0	0	0	0	0	0	0	0
Grenada	0	0	0	0	0	.001	0	0	0	0	0	0	0
Laos	0	.625	.101	1.000	.001	0	0	0	.225	.454	.039	0	0
Lesotho	0	0	.013	0	0	0	0	0	0	0	0	0	0
Mauritius	0	0	0	0	0	0	0	0	0	0	0	0	0
Morocco	0	0	0	0	0	0	0	0	0	0	0	0	0
Namibia	0	0	0	0	0	0	0	.068	0	0	0	0	0
Oman	0	0	0	0	0	0	0	0	0	0	0	.447	.043
Panama	0	0	0	0	0	0	0	0	0	0	0	0	0
Saint Kitts and N.	0	0	0	0	0	0	0	0	0	0	0	0	0
Saint Lucia	0	0	0	0	0	0	0	0	0	0	0	0	0
Seychelles	0	0	0	0	0	0	0	0	0	0	0	.068	0